

ETHAN TORRES

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Github - <https://github.com/Landbus>

EDUCATION

PhD from University of Illinois, Urbana-Champaign Industrial Engineering	Sep 2023 - 2027
MS from University of Illinois, Urbana-Champaign, Industrial Engineering	Sep 2023 - 2026
MS from University of Illinois, Urbana-Champaign, Applied Mathematics, Concentration in Computational Science and Engineering	Sep 2024 - 2025
BA from New York University Mathematics	Sep 2019 - 2023

HONORS AND AWARDS

Title of Award Deans List, NYU	2020
Title of Fellowship Jerry S. Dobrolovny	2023 -

RESEARCH EXPERIENCE

Dissertation , In Progress Advisor: Richard B. Sowers Shaped Curricula for Formation Control and Collision Avoidance in Multi Agent Reinforcement Learning (in progress - to be submitted January)	2023 -
Current Research , Completed, In Submission Co-Authors: Ramavarapu Sreenivas, Richard Sowers • On the Enumeration of All Unique Paths of Recombining Trinomial Trees. preprint available: https://arxiv.org/abs/2510.02727	2023 - 2025

Current Research, In Progress 2025 -
Co-Authors: Ryan Roach
● Selective Policy Retraining

Current Research, In Progress 2025 -
Co-Authors: None
● Subdominant Lyapunov Exponents Dictate Training

TEACHING EXPERIENCE

University of Illinois, Urbana-Champaign, Champaign May 2023 -
Teaching Assistant, Department of Industrial Engineering
● Courses Taught
○ SE-100 Introduction to ISE
○ IE-525 Stochastic Calculus
○ IE-525 Numerical Methods in Finance
○ IE-434 Deep Learning
● SE 100 Responsibilities - Grading, Lecturing
● IE 525 Responsibilities - Grading, Lecturing, Assisting PhD and Advanced Masters students with course content
● IE 434 Responsibilities - Same as above

RELEVANT COURSEWORK

Financial Accounting
Multivariable Calculus
Linear Algebra
Discrete Mathematics
Ordinary Differential Equations
Partial Differential Equations
Mathematical Finance
Theory of Probability
Complex Analysis
Numerical Analysis
Real Analysis
Abstract Algebra
Measure Theory
Fourier Analysis
Harmonic Analysis
Applied Stochastic Processes
Stochastic Calculus
Asymptotic Analysis of ODEs
Optimization of Large Systems
Linear Programming

Financial Computing
Dynamical Systems I
Statistical Learning Theory
Numerical Methods for PDEs
Stochastic Control Theory
Structure of Graphs
Deep Learning
Stochastic Combinatorics
Functional Analysis
Deep Generative Models
Geometric Control Theory
High-Dim Information Theory
Dynamical Systems in NNs
Computational Statistics/Non-Linear Programming
Quantum Machine Learning

PROFESSIONAL AFFILIATIONS

Devlin Energy, Summer 2022
Business Data Analyst - Designed the architecture for the companies CRM, performed ad hoc analysis of Solar Energy Data

LANGUAGES

English: Native Language

Spanish: Intermediate Listener, Intermediate Speaker, Advanced Reading and Writing

COMPUTER SKILLS

Programming: C++, Python, Excel, SQL, R

REFERENCES

Can provide more references, please contact me if needed.

Dr. Richard B. Sowers, [Professor]
[Mathematics, Statistics, Industrial Engineering]
[University of Illinois, Urbana-Champaign]
[104 S Mathews Ave, Urbana, IL 61801]
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[Industrial Engineering]

[University of Illinois, Urbana-Champaign]
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Dr. Jalal Shatah, [Professor, Department Head]
[Mathematics]
[New York University]
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Thomas Janowski, [Manager]
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